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CASE REPORT

Forensic identification of a fatal snakebite from Bungarus multicinctus (Chinese krait) by pathological and toxicological findings: a case report

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Abstract
Bungarus multicinctus (*B. multicinctus*) is one of the top ten venomous snakes in China, ranking first in lethality at 26.9–33.3%. However, to our knowledge, no forensic autopsy-related cases of death from *B. multicinctus* bite poisoning have been reported. There are surprisingly few reported cases of death from poisoning by other species of neurotoxic snakes. Neurotoxic snake venom is often highly toxic, and death can quickly occur when bitten in the wild if victims are not taken to a doctor in time. We presented a case of an adult female in Fujian Province of China who was bitten by a poisonous snake while digging for bamboo shoots in the mountains and died from the bite of *B. multicinctus* confirmed by enzyme-linked immunosorbent assays (ELISA) results. The autopsy's results, histopathological findings, and ELISA results reported here can be helpful for future forensic practice in *B. multicinctus* venom poisoning; we also briefly review the pathological changes of neurotoxin poisoning, which may be useful in other types of neurotoxin snake venom poisoning.

Keywords Forensic pathology · Forensic toxicology · Snakebite · *Bungarus multicinctus* · ELISA

Introduction

Venomous snake bite is a significant public health problem in developing countries. According to the World Health Organization estimates, approximately 5.4 million people are bitten by snakes worldwide each year, with the majority occurring in Asia, Africa, and Latin America, and up to 2 million people are bitten by snakes in Asia alone each year [1]. *Bungarus multicinctus* (*B. multicinctus*) is one of China's top ten venomous snakes. Although *B. multicinctus* bites are in the fifth position among all venomous snake bites in China, accounting for only 8.12%, it has the highest lethality ranking of 26.9–33.3% [2–4]. Due to climatic characteristics and geographical location, *B. multicinctus* is mainly distributed in subtropical regions of China, such as Guangdong, Guangxi, Fujian, and other provinces [5].

As *B. multicinctus* bites are rare, to our knowledge, no forensic autopsy cases of death by poisoning from *B. multicinctus* bites have been reported. According to the reported proteomics results, *B. multicinctus* venom is mainly composed of α-bungarotoxin (α-BGT), β-bungarotoxin (β-BGT), κ-bungarotoxin (κ-BGT), and γ-bungarotoxin (γ-BGT) [4, 6, 7]. Post-synaptic neurotoxins (i.e., α-BGT) and pre-synaptic PLA2 neurotoxins (i.e., β-BGT) play a significant

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toxicological role in the envenomation process by *B. multicinctus*. *B. multicinctus* injects venom into subcutaneous tissue through the proteroglyphic teeth, which is absorbed mainly by the lymphatic system and then enters the blood circulation for distribution throughout the body [7]. The neurotoxins of *B. multicinctus* mainly block the conduction of motor neuromuscular junction, leading to flaccid paralysis of the striated muscles, including respiratory muscles, and the victim eventually dies of acute respiratory failure [7].

We presented a case of an adult female in Fujian Province of China who was bitten by a poisonous snake while digging for bamboo shoots in the mountains and died from the bite of *B. multicinctus* confirmed by ELISA results.

Case report

Case history

One day in May, Fujian Province of China, a woman met her friends at 05:00 to dig bamboo shoots in the mountains. Around 10:00, she left the group and entered the deep mountains alone. She was missing when her friends were ready to go home around 11:00. At 14:00, her relatives went into the mountains to search for her and failed. Then, her husband called the police for help. The police organized a rescue team to search the mountain at 06:00 the next day and found the woman's body at around 09:00. The timeline is shown in Fig. 1a. The distance between the group location, the missing location, and the body location is illustrated in Fig. 1b. Horizontal line distance between the group location and the missing location is 204 m. Horizontal line distance between the missing location and the body location is 688 m. Due to the slope of the mountain and other geographical reasons, the actual distance is more likely to be further. The body was found in a semi-recumbent position with face down (Fig. 1c).

Autopsy finding

The autopsy was performed within 24 h. Conjunctival petechiae were observed bilaterally. The lips and fingernails were visibly cyanosis. Scattered abrasions and contusions were seen on the face and extremities of the body. A small amount of yellowish regurgitated gastric contents was seen in the trachea (Fig. 2a).

Suspected snake bites were found on the dorsum of the right foot and left wrist. Two pinpoint-like defects were found on the dorsal skin of the right foot, 1.0 cm apart, with dark red surface, and subcutaneous hemorrhage was seen in the cut section (Fig. 3a, b). Two pinpoint-like defects were found on the skin of the left wrist, 0.9 cm apart, with redness surface and subcutaneous edema were seen in the cut section (Fig. 3c, d).

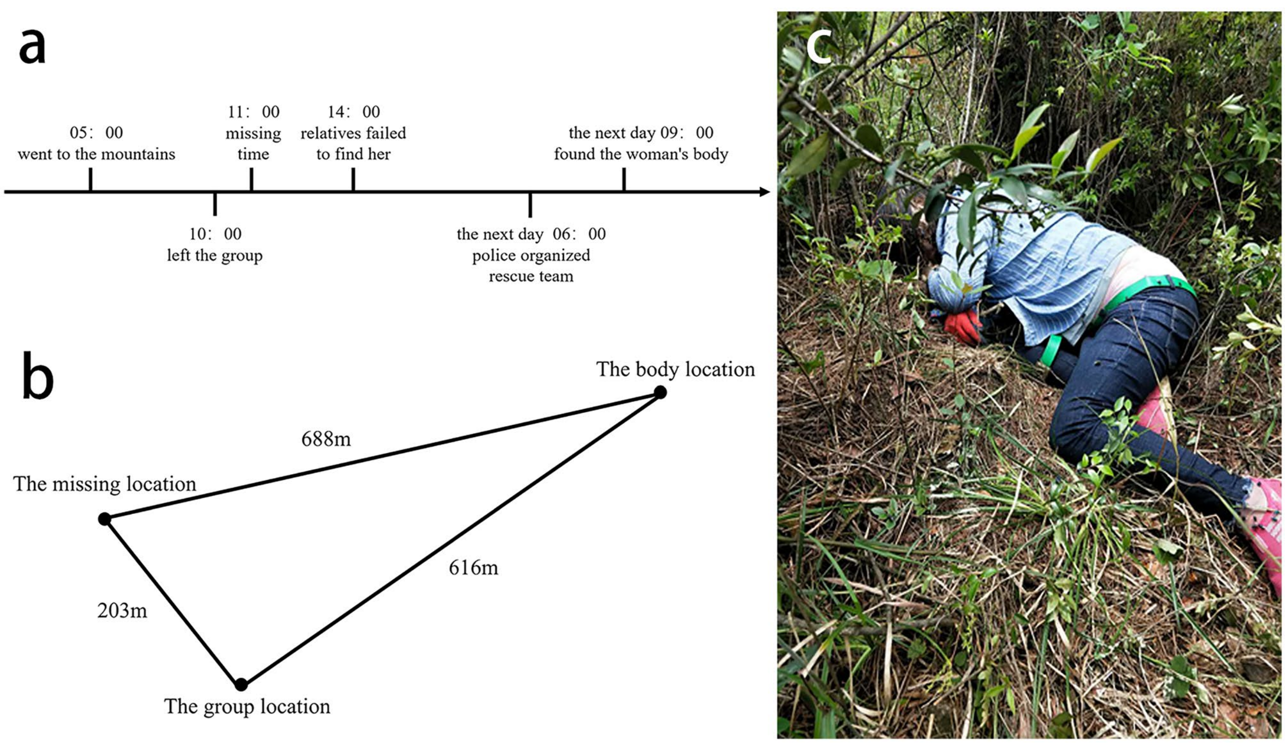

Fig. 1 **a** The timeline in this case. **b** The distance between the group location, the missing location, and the body location. **c** Photograph of the scene, the body was found in a semi-recumbent position with face down

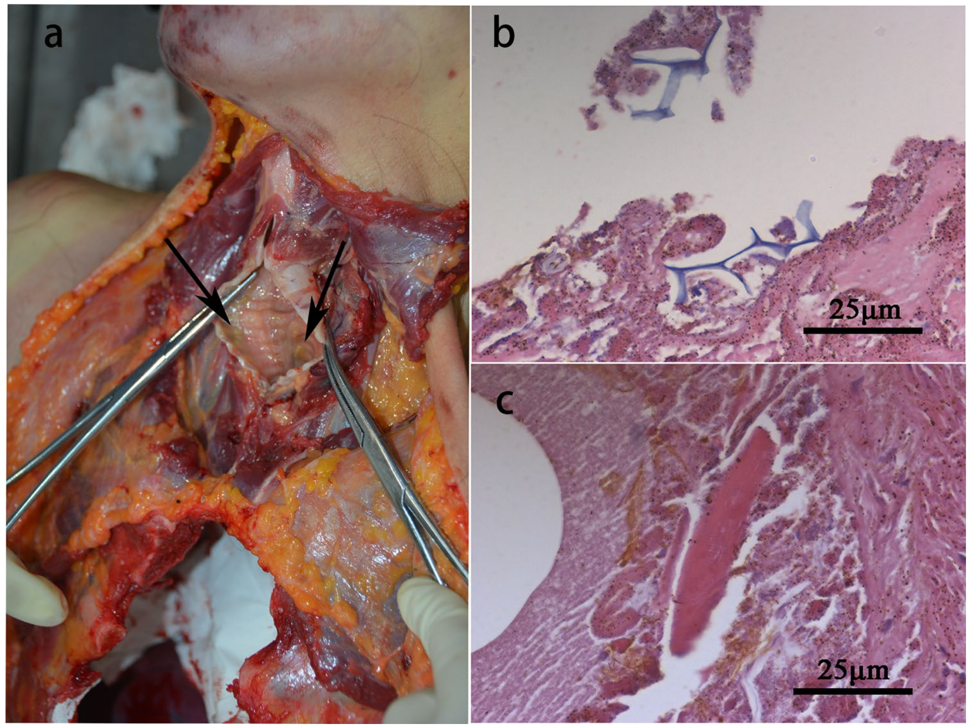

Fig. 2 **a** Small amount of yellowish regurgitated gastric contents seen in the victim's trachea (↘). **b** Suspected plant cell wall in the lung tissue. **c** Suspected muscle fibers in the lung tissue

Pathologic findings

Microscopic revealed interrupted epidermal continuity of the skin at the suspected snake bite on the left wrist and scattered small patchy hemorrhages in the subcutaneous soft tissue. The skin of the suspected snake bite on the dorsum of the right foot was seen to have interrupted epidermal continuity and a large patchy of subcutaneous soft tissue hemorrhage. No tissue necrosis was found either.

Suspected muscle fibers and plant cell walls were discovered in the part of bronchiole cavity and alveoli cavity (Fig. 2b,

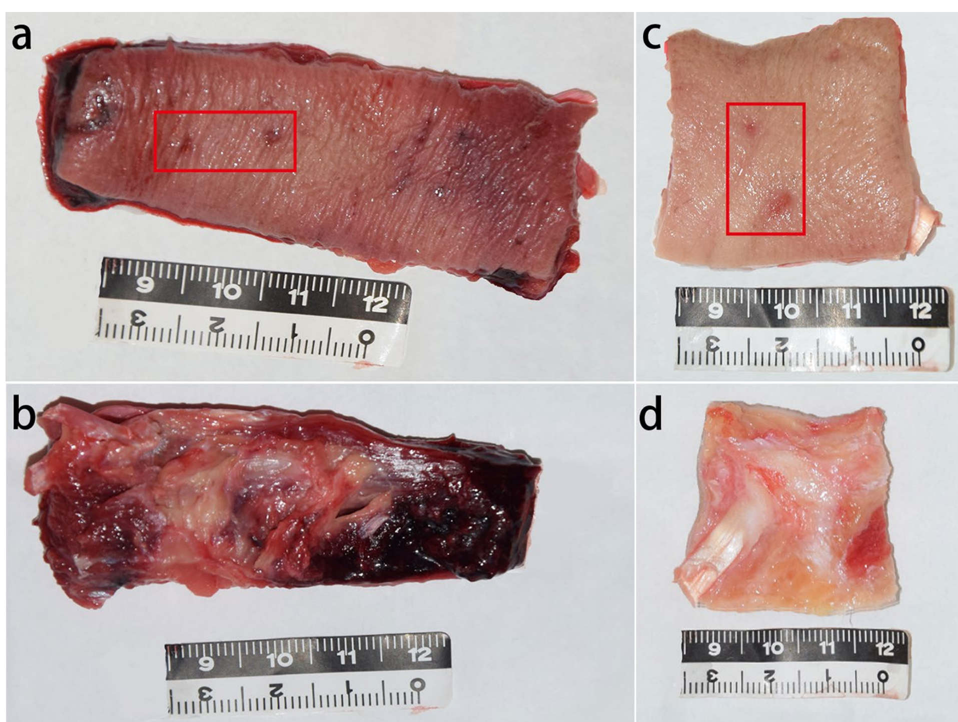

Fig. 3 **a** Front side of the skin of the suspected snake bite on the dorsum of the right foot. Suspected snake bite marks (☐). **b** Back side of the skin of the suspected snake bite on the dorsum of the right foot. **c** Front side of the skin of the suspected snake bite on the left wrist. Suspected snake bite marks (☐). **d** Back side of the skin of the suspected snake bite on the left wrist

c). Other pulmonary pathology includes pulmonary edema and occasional small patchy pulmonary hemorrhage.

The epicardium was found with scattered focal hemorrhage, fatty infiltration of the right ventricular myocardium, and coronary arteries are normal. Other pathological changes include mild cerebral edema, mild neuronal degeneration and necrosis, fatty changes in the liver, and partial glomerulosclerosis.

Tissue samples obtained at autopsy were frozen at −80 °C prior to sandwich-ELISA test.

Laboratory examination

Sandwich-ELISA test

Based on the geographical distribution of venomous snakes in China and the pathological changes observed [5], *Deinagkistrodon acutus* (*D. acutus*), *Agkistrodon halys*, and *B. multicinctus* were selected for the ELISA test. Sandwich-ELISA test was adapted according to methods described by Huang et al. [8]. Snake venom (all in powder) of *D. acutus*, *Agkistrodon halys*, and *B. multicinctus* were offered by Kunming Institute of Zoology, Chinese Academic of Science. *B. multicinctus* venom was detected in the skin of both the left wrist and the dorsum of the right foot at the suspected snake bite, 2.3 ng/ml of the left wrist and 3.4 ng/ml of the right foot. *B. multicinctus* venom was not detected in the urine.

Toxicology tests

Stomach contents and heart blood were extracted for toxicological testing, and no common toxicants were found.

Diagnosis

After autopsy and laboratory tests, it was shown that the adult female died of acute respiratory failure due to poisoning caused by *B. multicinctus* bite. Aspiration caused by regurgitation of the stomach contents blocking part of bronchiole cavity and alveoli cavity played a supporting role in the death process.

Discussion

Forensic identification of venomous snake cases requires a combination of the history, characteristic fang marks, and snake venom detection [8]. The present case occurred in May, just at the peak periods of *B. multicinctus* activity, which is consistent with the habits of *B. multicinctus* [7]. The victim disappeared at 11:00 and was found dead at 09:00 the next day, with an interval of 22 h. The median time of apnea after *B. multicinctus* bite poisoning was around 6 h [3], and the missing location was more than 688 m away from the body location, within 6 h of where the victim could move. *B. multicinctus* bite marks generally have two pinpoint-like size tooth marks, spacing 0.8 to 1.5 cm apart, and surrounding tissues are not obvious represent red or swollen [3]. The multiple snake bite marks of the victim in present case were consistent with the descriptions in the literature. Although there is hemorrhage at the dorsum of the right foot bite marks, no tissue necrosis was found. ELISA is currently the most widely used method for detecting snake venom [9]. In this case, sandwich-ELISA test detected *B. multicinctus* venom at the bite marks on the victim's left wrist and the dorsum of the right foot, and the cause of death was confirmed to be acute respiratory failure from *B. multicinctus* bites.

Unlike other species of venomous snakes, the amount of venom excreted by *B. multicinctus* is minimal, only 4.6 mg, and the lethal dose for adults is about 1 mg [3]. The low quantity of venom excreted by *B. multicinctus* may mean its venom detection could be challenging. In this case, the subcutaneous tissue of the suspected snake bite mark was selected for ELISA, and a small concentration of *B. multicinctus* venom was detected. Although there are studies to detect *B. multicinctus* venom in urine [9, 10], the urine was negative in present case, which may attributed to postmortem putrefaction and degradation of venom proteins. This suggests that snake venom testing should be performed as early as possible in cases of *B. multicinctus* venom poisoning, and the subcutaneous tissue of snake bite is a better optimal test sample than urine.

The examination of neurotoxin poisoning remains a sticking point in forensic practice. Neurotoxins affect the conduction of motor neuromuscular junction resulting in blepharoptosis, dysphagia, myasthenia of limbs, and other clinically manifestations in the early stage [11]. However, only nonspecific signs could be found in autopsy and pathological examinations. Summarizing the case reports of deaths from this and other types of neurotoxin snake venom poisoning reveals the following commonalities [12–14]: ① Pinpoint-like snake bite marks are often found on the corpse surface (especially on the extremities) with rather normal surrounding tissues. ② Petechial hemorrhage is produced by increased permeability of small vessels due to hypoxia found in the conjunctiva of the eyelids, pulmonary pleura, epicardium, etc. [15]. ③ In the central nervous system, the direct effects of neurotoxins need to be further investigated [11]. In this case, only mild neuronal degeneration and necrosis and mild cerebral edema due to hypoxia were found. ④ The effect of neurotoxins on the neuromuscular junction is mainly functional, and there is no significant necrosis in muscle tissue [7]. The muscle tissue in this case also showed no apparent signs of necrosis. ⑤ There is a case report of significant nephrotoxicity in victims bitten by krait [12], but no toxic effects of *B. multicinctus* venom on other organs have been identified [11]. In this case, the victim's liver and kidney showed pathological changes of their own disease.

Neuromuscular blockade due to neurotoxin is similar to residual neuromuscular blockade during anesthesia. It increases the risk of airway obstruction and aspiration due to muscle weakness and ataxia, impairing the function of the pharynx and upper airway [16]. When aspiration occurs, foreign objects such as gastric contents can be found under the microscopical examination of trachea and lung tissues. In this case, the victim was found in a semi-recumbent position with her face down, further contributing to the aspiration. A small amount of yellowish gastric contents was found in the victim's trachea, but the airway was not obstructed, while gastric contents such as plant cell walls were found in the bronchiole cavity and alveolar cavity, which is consistent with the phenomenon of aspiration. Aspiration of a small amount of gastric contents from the upper airway was also found in another case report of neurotoxin snake venom death [13]. This indicates that aspiration in cases of neurotoxin snake venom poisoning may have a potential specific value. At the same time, aspiration was also associated with many factors such as death posture and gastroesophageal reflux diseases during the agonal stage. In forensic practice, aspiration should be judged concretely with the circumstances of the case whether it is related to neurotoxins.

In conclusion, to our knowledge, no forensic autopsy-related cases of death from *B. multicinctus* bite poisoning have been reported. There are surprisingly few reported cases of deaths from poisoning by other species of neurotoxic snakes, which means it is a neglected area. Forensic practice encountered corpses in the mountains and forests; when autopsy could not discover obvious positive signs, consideration should be given to death of neurotoxic snake. The corpse's surface (especially the extremities) should be carefully examined for signs of suspected snake bites and whether aspiration occurs in the trachea and pulmonary tissue. We hope that the autopsy results, histopathological findings, and ELISA results reported here can be helpful for future forensic practice in *B. multicinctus* venom poisoning and other types of neurotoxin snake venom poisoning.

Key points

1. To our knowledge, no forensic autopsy-related cases of death from B. multicinctus bite poisoning have been reported.
2. We report a case of B. multicinctus bite poisoning with autopsy results, histopathological findings and ELISA results.
3. The pathological changes of neurotoxin snake poisoning was discussed.
4. Forensic practice encountered corpse in wild, when autopsy could not discover obvious positive signs, consideration should be given to death of neurotoxic snake poisoning.
5. The phenomenon of aspiration in neurotoxin snake poisoning may have a potential specific value.

Declarations

Informed consent Written informed consent was obtained from the family members of the patient.

Conflict of interest The authors declare thatthey have no conflict of interest.

References

1. World Health Organization. Snakebite envenoming. https://www.who.int/news-room/fact-sheets/detail/snakebite-envenoming. Accessed 17 May 2021.
2. Mao YC, Liu PY, Chiang LC, Liao SC, Su HY, Hsieh SY, et al. Bungarus multicinctus multicinctus snakebite in Taiwan. Am J Trop Med Hyg. 2017;96:1497–504. https://doi.org/10.4269/ajtmh.17-0005.
3. Li HJ, Li GM, Li PR, Zhao W. Progress in the treatment of Bungarus multicinctus bites and research. J Snake. 2007:205–8. https://doi.org/10.3969/j.issn.1001-5639.2007.03.015.
4. Liang Q, Huynh TM, Ng YZ, Isbister GK, Hodgson WC. In vitro neurotoxicity of Chinese krait (Bungarus multicinctus) venom and neutralization by antivenoms. Toxins. 2021;13:49. https://doi.org/10.3390/toxins13010049.
5. Hu SZ, Yang H, Zhang J, Tian JP, Liu ZH. Species conservation and resource development of the Bungarus multicinctus. In: 2006 Cross-Strait and CSNR National Seventh Symposium on Natural Drug Resources. Wuhan, Hubei, China. 2006:128–132.
6. Shan LL, Gao JF, Zhang YX, Shen SS, He Y, Wang J, et al. Proteomic characterization and comparison of venoms from two elapid snakes (Bungarus multicinctus and Naja atra) from China. J Proteom. 2016;138:83–94. https://doi.org/10.1016/j.jprot.2016.02.028.
7. Deng HX, Li QB. Research progress on the pathogenesis and treatment of Bungarus multicinctus bite poisoning. J Snake. 2008:35–46. https://doi.org/10.3969/j.issn.1001-5639.2008.01.018.
8. Huang F, Zhao S, Tong F, Liang Y, Le Grange JM, Kuang W, et al. Unexpected death in a young man associated with a unilateral swollen leg: pathological and toxicological findings in a fatal snakebite from Deinagkistrodon acutus (Chinese moccasin). J Forensic Sci. 2021;66:786–92. https://doi.org/10.1111/1556-4029.14622.
9. Gao JF, Wang J, Qu YF, Ma XM, Ji X. Immunoreactivity between venoms and commercial antiserums in four Chinese snakes and venom identification by species-specific antibody. J Immunol Methods. 2013;387:211–8. https://doi.org/10.1016/j.jim.2012.10.016.
10. Dong le V, Selvanayagam ZE, Gopalakrishnakone P, Eng KH. A new avidin-biotin optical immunoassay for the detection of beta-bungarotoxin and application in diagnosis of experimental snake envenomation. J Immunol Methods. 2002;260:125–36. https://doi.org/10.1016/s0022-1759(01)00527-0.

11. Ranawaka UK, Lalloo DG, de Silva HJ. Neurotoxicity in snakebite–the limits of our knowledge. PLoS Negl Trop Dis. 2013;7:e2302. https://doi.org/10.1371/journal.pntd.0002302.
12. Sarkar N, Basu S, Chandra P, Chowdhuri S, Mukhopadhyay PP. Nephrotoxicity in krait bite: a rare case series of three fatalities in consecutive bites by a single snake. Egypt J Forensic Sci. 2018;8. https://doi.org/10.1186/s41935-018-0040-3.
13. Norris RL, Pfalzgraf RR, Laing G. Death following coral snake bite in the United States–first documented case (with ELISA confirmation of envenomation) in over 40 years. Toxicon. 2009;53:693–7. https://doi.org/10.1016/j.toxicon.2009.01.032.
14. Silva A, Gamlaksha D, Waidyaratne D. Medico-legal significance of the identification of offending snake in a fatal snake bite: a case report. J Forensic Leg Med. 2013;20:965–7. https://doi.org/10.1016/j.jflm.2013.09.009.
15. Necas P, Hejna P. Eponyms in forensic pathology. Forensic Sci Med Pathol. 2012;8:395–401. https://doi.org/10.1007/s12024-012-9328-z.
16. Murphy GS, Brull SJ. Residual neuromuscular block: lessons unlearned. Part I: definitions, incidence, and adverse physiologic effects of residual neuromuscular block. Anesth Analg. 2010;111:120–8. https://doi.org/10.1213/ANE.0b013e3181da832d.